SMART BUS SYSTEM FOR REAL TIME PASSENGER MANAGEMENT SYSTEM

SLIIT
FACULTY OF COMPUTING

Meet our team

Weerakoon W.M.D.P.

IT22131560

Oshadha Pathiraja

IT22230492

Vidumini Chandrasekara

IT22286246

Abesekara W R G M

IT22271150

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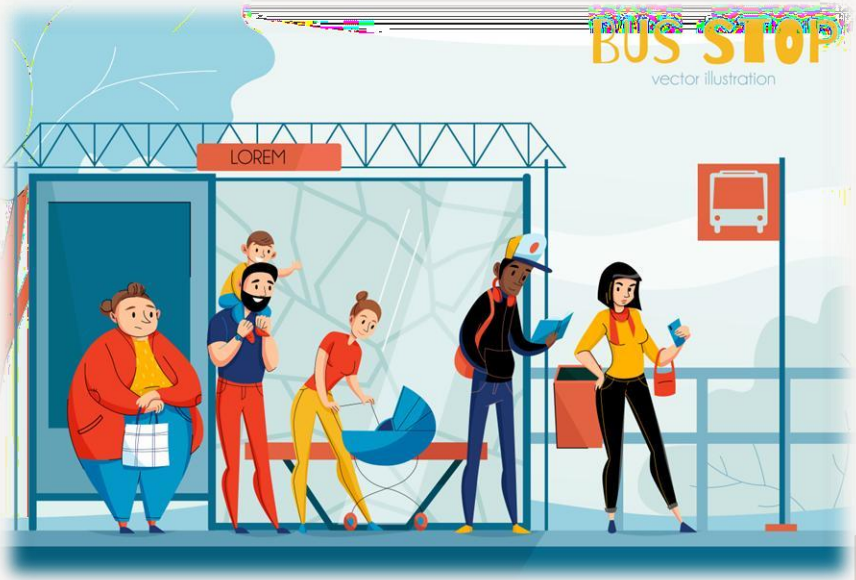
Problem and It's Impact

- Public transport is critical in Sri Lanka, with approximately 1.45 million daily passengers relying on SLTB buses.
- The current system suffers from **unpredictable schedules**, severe **overcrowding** during peak hours, and a lack of real-time information, causing passengers to miss buses or face unsafe travel conditions.
- There is currently no reliable method for passengers to check seat availability or communicate needs, and no data-driven way for authorities to manage fleet efficiency.

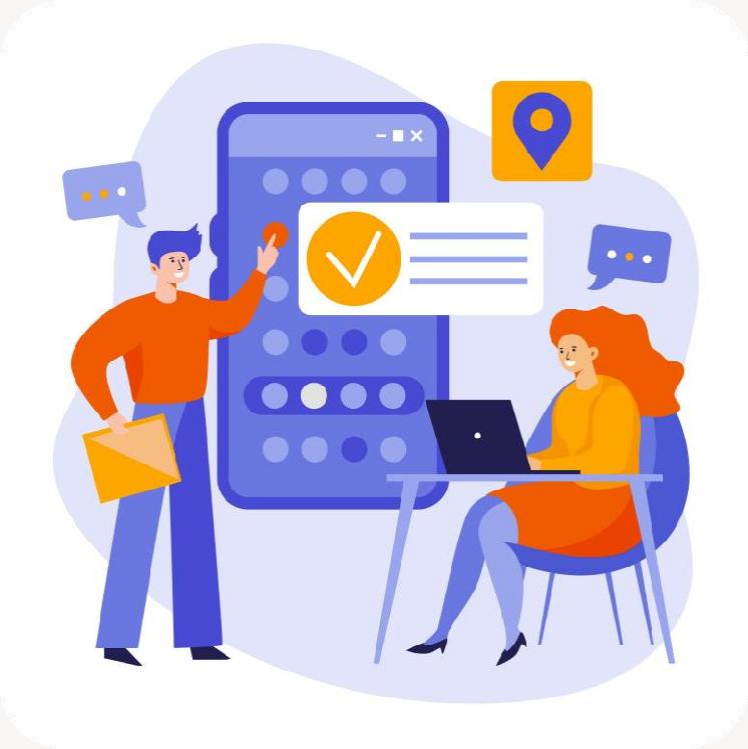
Who will be benefited

- Daily Commuters
- Transport Authorities (SLTB) And Private Bus owners
- Travelers who are new to the area



Design Excellence

- Use IoT devices for get accurate Location, Weather data, and passenger count
- Design a user-friendly app.
- Provide functionalities to driver to manage passengers.
- Manage all the buses with admin dashboard



IT22131560 | WEERAKOON W.M.D.P

IoT Based Bus Tracking System And ETA Prediction

Specialization: Information Technology

Project ID: 25-26J-267

Problem

- Passengers often face **uncertainty in bus arrival times**.
- If we can predict accurate arrival time of bus, it will save so much time of public tranceport users.

Solution

- Designed IoT Based ETA Prediction system to get more accurate arrival time to user location.
- Designed user friendly interface to interact easily with the app.

User Requirements Addressed by The Solution

- Accurate Arrival Time Prediction
- User friendly App
- Multiple Options to check what is fastest option.
- Intedgrated Map to track live location

Design Exelence

- Existing bus tracking solutions are **not accurate enough**, as they don't consider all real-world conditions.
- Here we get accurate GPS data. Weather data. And traffic data from google to get more accurate arrival time prediction.
- Consider average passenger onboarding time.

User Feedback

- Can easily interact with interface. And much needed details are available
- Bus driver is ok with connecting mobile hotspot to the device.

IT22230492 | PATHIRAJA H.P.M.O.N.

Central Fleet Management and Route Optimization Engine

Specialization: Information Technology

Project ID: 25-26J-267

Problem

- Passengers experience unexpected delays due to inaccurate ETA for their destination
- Lack of real-time predictions for buses reach passenger's destination
- Lack of transparency in ticket pricing.
- No real-time system to report driver behavior, reducing passenger safety.

Solution

- Mobile UI to show prediction results after entering “From” and “To” destinations
- Using real-time IoT data bus and Google Maps API.
- Pre-journey ticket fare estimation based on approved pricing rules.
- Real-time passenger complaint system with driver alerts.

User Requirements Addressed

- Ability to view accurate, real-time ETA predictions for selected destinations.
- Access to live bus tracking and arrival predictions using real-time IoT data.
- Transparent ticket fare estimation before starting the journey.
- Facility to submit real-time complaints regarding driver behavior

Design Excellence

- Responsive mobile UI design to improve passenger UX and user interaction.
- Minimal user input to reduce complexity.
- Clear and very accruable prediction output
- Scalable architecture supporting multiple routes and time ranges for bus operations.
- Future-ready design enabling passenger-to-driver messaging and complaint handling.
- Future integration of transparent ticket fare calculation based on government regulations.

User Feedback

- Accurate ETA predictions improved journey planning
- Simple and responsive UI enhanced user experience
- Fare estimation increased ticket price transparency
- Real-time complaints improved passenger safety

IT22286246 | CHANDRASEKARA C.M.P.V

Mobile-Based Passenger Interaction System

Specialization: Information Technology

Project ID: 25-26J-267

Problem

- Passengers face uncertainty about bus arrival time
- Overcrowding causes discomfort, delays, and safety concerns.
- Lack of real-time communication between passengers and drivers.
- No crowd forecast for better journey planning.

Solution

- Mobile-based ML crowd prediction system
- Simple output messages for decision making
- Passenger-to-driver messaging feature

User Requirements Addressed

- Predict crowd before travel
- Simple mobile interface
- Clear and understandable travel guidance

Design Excellence

- User-centered mobile UI designed for daily bus commuters
- Minimal user inputs to reduce complexity
- Clear and actionable prediction output
- Scalable architecture to support multiple routes and time ranges
- Future-ready design enabling passenger-to-driver messaging integration

User Feedback

- UI is easy to understand
- Useful for daily travel planning

IT22271150 | Abeysekara W.R.G.M

Vision based passenger counting & Occupancy

Specialization: Information Technology

Project ID: 25-26J-267

Problem

- Users cannot know how many seats available before boarding.
- Overcrowding causes discomfort, delays, and safety concerns.
- Lack of real-time seat occupancy and crowd checking.
- Authorities make decisions based on guesses instead of real data.

Solution

- Two cameras at bus entry and exit doors automatically detect boarding and alighting passengers.
- Mobile app shows current passenger count before and during travel.
- Authorities receive accurate passenger data to optimize routes and schedules.

User Requirements Addressed

- Predict crowd before travel
- Simple mobile interface
- Real time updates

Design Excellence

- Real time system architecture
- Two-camera accuracy enhancement
- Clear and actionable prediction output
- Scalable modular architecture

User Feedback

- The app helps me decide whether to board the bus or wait for the next one.
- The app interface is simple and easy to understand.

Commercialization

Customer persona

- **Daily Commuters:** The 1.45 million daily passengers who face uncertainty and overcrowding.
- **Transport Authorities (SLTB/Private):** Fleet managers seeking data-driven decisions to optimize resources

Market Size

- Approximately **20,000** buses are available for public transportation in Sri Lanka
- **4M - 5M people** use public transport buses daily.

How will the cost be recovered?

- Sell This system for SLTB and Private bus companies.
- SLTB or Other private companies can use this system for gain customers and increase Profit

Thank You!